hw_class06

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Section 1

A.

> Improve this regular R code by abstracting the main activities in your own new function.

```
df <- data.frame(a=1:10, b=seq(200,400,length=10),c=11:20,d=NA)
df$a <- (df$a - min(df$a)) / (max(df$a) - min(df$a))
df$b <- (df$b - min(df$a)) / (max(df$b) - min(df$b))
df$c <- (df$c - min(df$c)) / (max(df$c) - min(df$c))
df$d <- (df$d - min(df$d)) / (max(df$a) - min(df$d))
```

Improvement

```
dffunction <- function(x) {
  (x - min(x)) / (max(x) - min(x))
}
```

```
df <- data.frame(a=1:10, b=seq(200,400,length=10),c=11:20,d=NA)
df$a <- dffunction(df$a)
df$b <- dffunction(df$b)
df$c <- dffunction(df$c)
df$d <- dffunction(df$d)

df
```

```
           a         b         c  d
1  0.0000000 0.0000000 0.0000000 NA
2  0.1111111 0.1111111 0.1111111 NA
3  0.2222222 0.2222222 0.2222222 NA
4  0.3333333 0.3333333 0.3333333 NA
5  0.4444444 0.4444444 0.4444444 NA
6  0.5555556 0.5555556 0.5555556 NA
7  0.6666667 0.6666667 0.6666667 NA
8  0.7777778 0.7777778 0.7777778 NA
9  0.8888889 0.8888889 0.8888889 NA
10 1.0000000 1.0000000 1.0000000 NA
```

B.

> Next improve the below example code for the analysis of protein drug interactions by abstracting the main activities in your own new function.

```
library(bio3d)
s1 <- read.pdb("4AKE") # kinase with drug
```

```
  Note: Accessing on-line PDB file
```

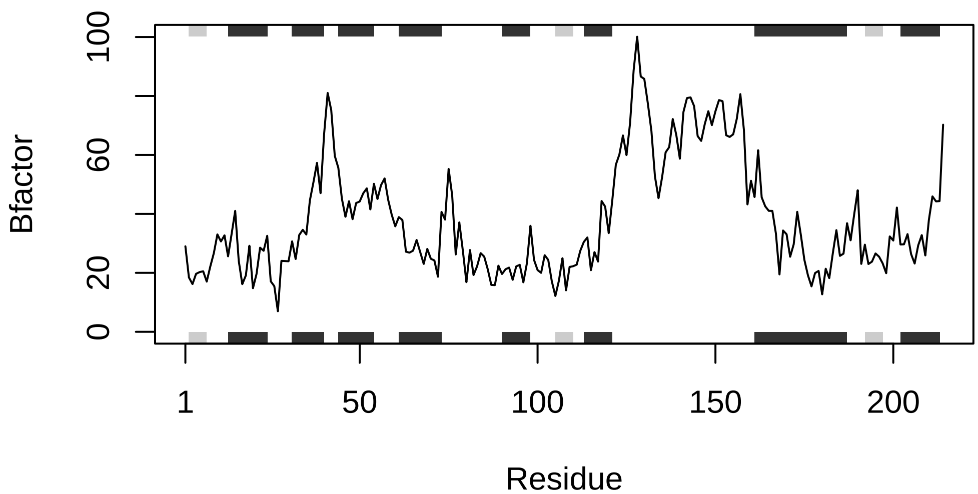
```
s2 <- read.pdb("1AKE") # kinase no drug
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

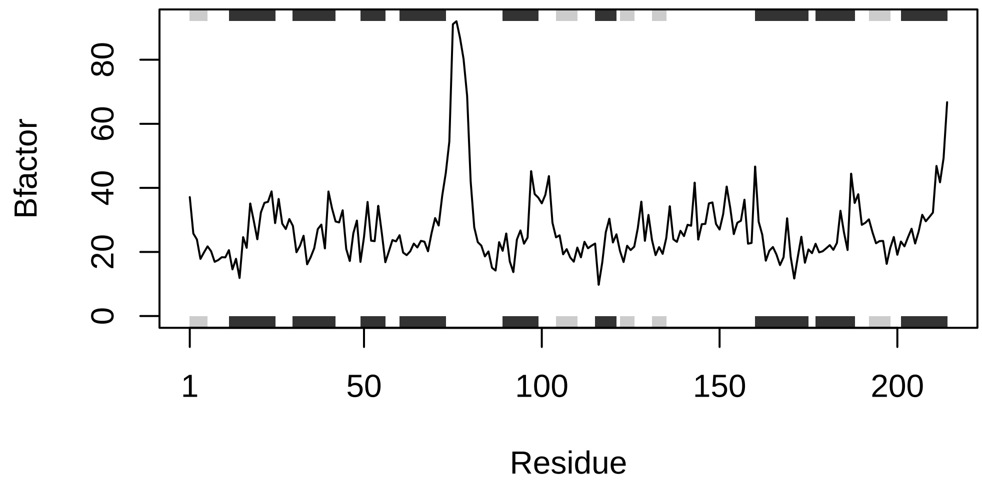
```
s3 <- read.pdb("1E4Y") # kinase with drug
```

```
  Note: Accessing on-line PDB file
```

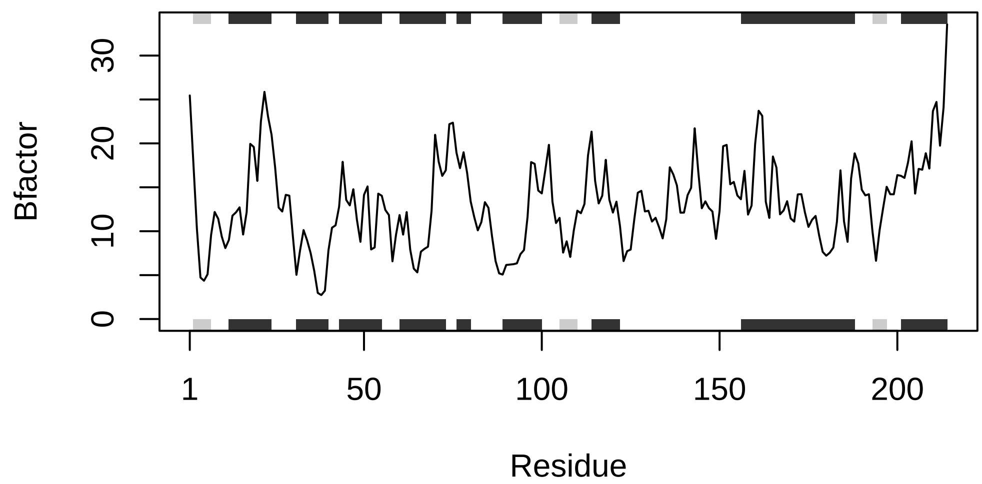
```
s1.chainA <- trim.pdb(s1, chain="A", elety="CA")
s2.chainA <- trim.pdb(s2, chain="A", elety="CA")
s3.chainA <- trim.pdb(s3, chain="A", elety="CA")
s1.b <- s1.chainA$atom$b
s2.b <- s2.chainA$atom$b
s3.b <- s3.chainA$atom$b
plotb3(s1.b, sse=s1.chainA, typ="l", ylab="Bfactor")
```




```
plotb3(s2.b, sse=s2.chainA, typ="l", ylab="Bfactor")
```




```
plotb3(s3.b, sse=s3.chainA, typ="l", ylab="Bfactor")
```



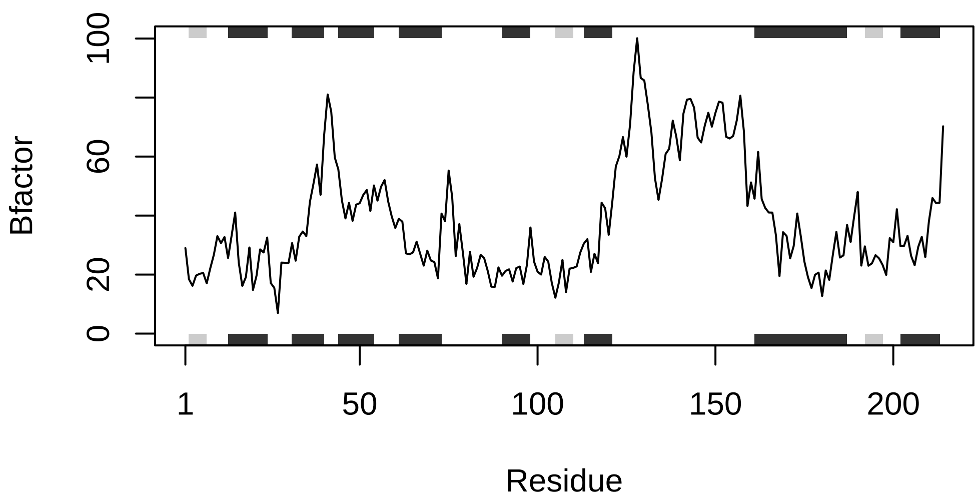
Improvement

```r
bio3dfunction <- function(id) {
  pdb <- read.pdb(id)
  pdb_chainA <- trim.pdb(pdb, chain="A", elety="CA")
  pdb_b <- pdb_chainA$atom$b
  plotb3(pdb_b, sse=pdb_chainA, typ="l", ylab="Bfactor")
}
```

```r
bio3dfunction("4AKE")
```

```
  Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/23/2nv742v14zl2lf7_lyd6kmqc0000gn/T//RtmpmRcf7r/4AKE.pdb exists.
Skipping download
```

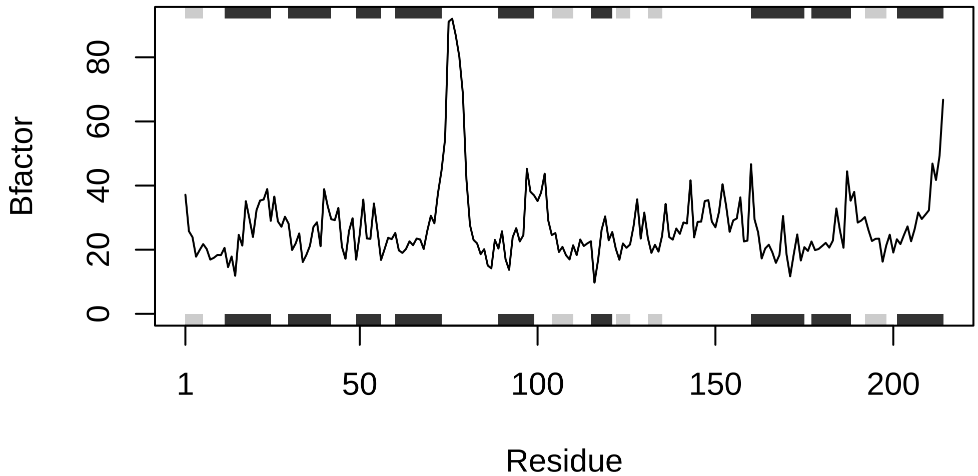



```r
bio3dfunction("1AKE")
```

```
  Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/23/2nv742v14zl2lf7_lyd6kmqc0000gn/T//RtmpmRcf7r/1AKE.pdb exists.
Skipping download
```

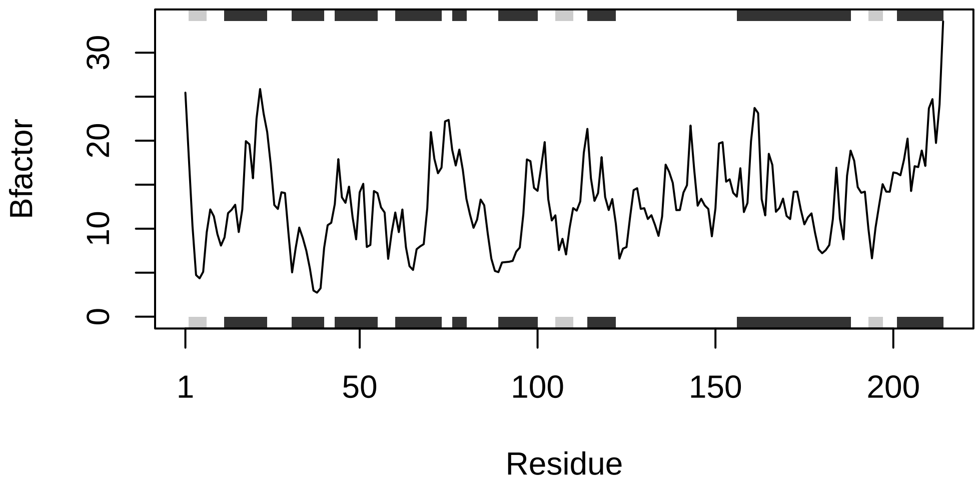
```
PDB has ALT records, taking A only, rm.alt=TRUE
```




```r
bio3dfunction("1E4Y")
```

```
  Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/23/2nv742v14zl2lf7_lyd6kmqc0000gn/T//RtmpmRcf7r/1E4Y.pdb exists.
Skipping download
```



HW (Use code in B.)

Q6. How would you generalize the original code above to work with any set of input protein structures?

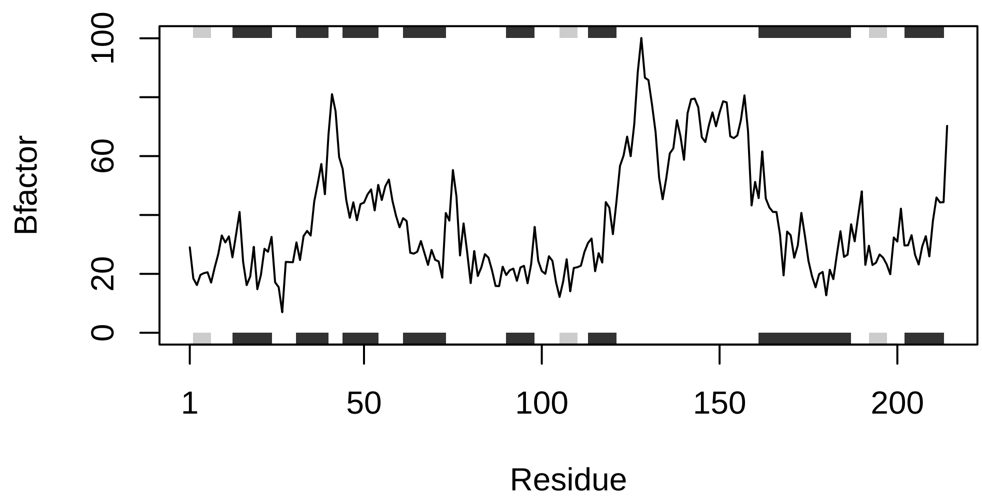
Write your own function starting from the code above that analyzes protein drug interactions by reading in any protein PDB data and outputs a plot for the specified protein.

```
bio3dfunction <- function(id) { # INPUT: pdb id
  pdb <- read.pdb(id) # read pdb file of specified id
  pdb_chainA <- trim.pdb(pdb, chain="A", elety="CA") # stores ONLY chain A and CA atoms from
  pdb_b <- pdb_chainA$atom$b # stores numeric vector to 'pdb_b'
  plotb3(pdb_b, sse=pdb_chainA, typ="l", ylab="Bfactor") # plot B-factor using numeric vecto
} # OUTPUT: final B-factor vs. Residue plot
```

```
bio3dfunction("4AKE")
```

```
  Note: Accessing on-line PDB file
```

```
Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/23/2nv742v14zl2lf7_lyd6kmqc0000gn/T//RtmpmRcf7r/4AKE.pdb exists.
Skipping download
```

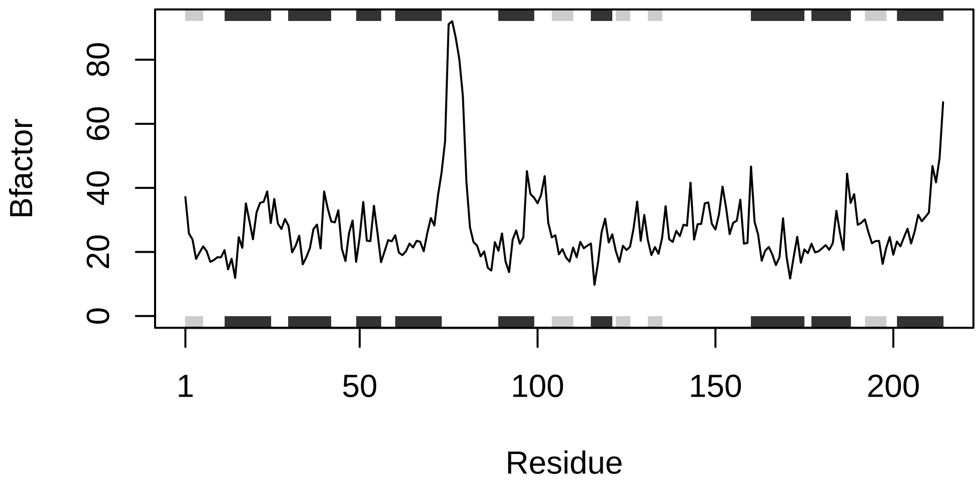



```r
bio3dfunction("1AKE")
```

```
  Note: Accessing on-line PDB file
```

```
Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/23/2nv742v14zl2lf7_lyd6kmqc0000gn/T//RtmpmRcf7r/1AKE.pdb exists.
Skipping download
```

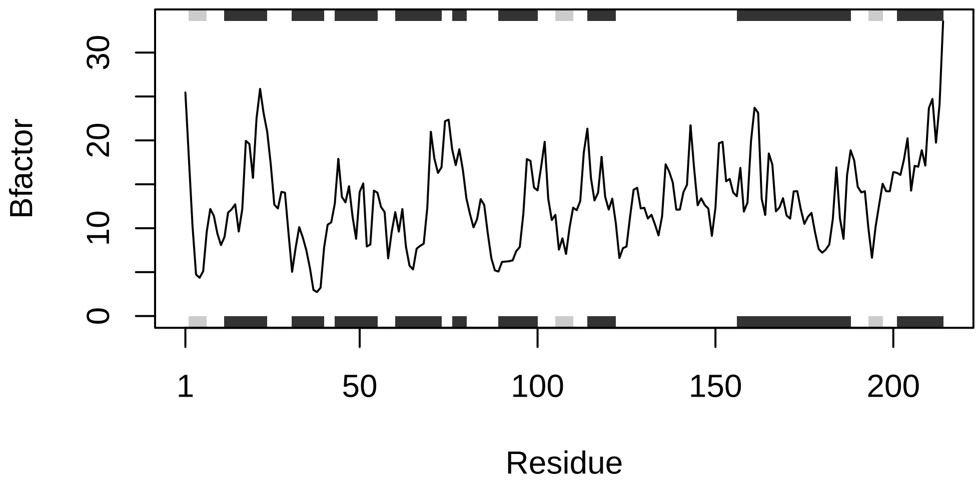
```
   PDB has ALT records, taking A only, rm.alt=TRUE
```




```r
bio3dfunction("1E4Y")
```

```
##   Note: Accessing on-line PDB file

## Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
## /var/folders/23/2nv742v14zl2lf7_lyd6kmqc0000gn/T//RtmpmRcf7r/1E4Y.pdb exists.
## Skipping download
```


Bfactor
0
10
20
30
1
50
100
150
200
Residue